TECHNICAL UNIVERSITY OF MUNICH

School of Computation, Information and Technology
Informatics

Master's Thesis in Informatics

DRAFT - Thesis Proposal: Extension of Pacing Diagram and Pacing Analysis in p:xe

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Thesis Proposal: Irgendetwas mit Usability in PIX:E

Author: Rai Trouvain
Supervisor: Prof. Dr. Georg Groh
Advisors: Julian Geheeb M.Sc.
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Disclaimer

I confirm that this Master's thesis is my own work and I have documented all sources and material used.

(Date) (Rai Trouvain)

Acknowledgements

The acknowledgements

Abstract

English Abstract

Zusammenfassung

German Abstract

Contents

1 Thesis Goal/Research Questions

The main goal of this thesis is to expand on the pacing diagram functionality in p:xe to capture and analyze more types of data that might be interesting in the video game design process.

- RQ1: Which data is interesting?
- RQ2: Which analysis of the data is interesting?

2 Approach

2.1 Original Pacemaker Functionality

After familiarizing myself with the existing codebase, the first step will be to port the original pacemaker functionalities. This includes basic path selection and corresponding intensity/gameplay category diagrams, as well as automatic calculation of some experience parameters (Geheeb, Dyrda, and Geheeb 2024).

2.2 Extend Diagram-Based Analysis

2.2.1 Path Selection

In the long run, the diagram should be extended to include path selection with concurrency and nestedness. While the implementation of this might not end up being part of my thesis, the software architecture and any changes I make should be extendable in that regard.

2.2.2 Lock/Key Dynamic

One big feature will be the analysis of lock/key dynamics and skills requirements. The basis of this will be the incorporation of locks/keys in the existing diagrams and the path selection functionality. The latter requires implementation of a specialized pathfinding algorithm like the one proposed by Aversa, Sardina, and Vassos (2015). The path selection would already offer a feasibility test, i.e. whether the lock/key puzzle is even solvable. Other approaches for useful analyses might be a challenge/entertainment rating (Font et al. 2016) or similarity checks that could help prevent potentially boring repetitions.

2.2.3 LLM-based evaluation

As pix:e already includes some LLM-based features, it could be easily extended to offer LLM-based evaluations along paths. Some ideas for this kind of analysis would be a narrative summary or the extraction of experience parameters from beat descriptions.

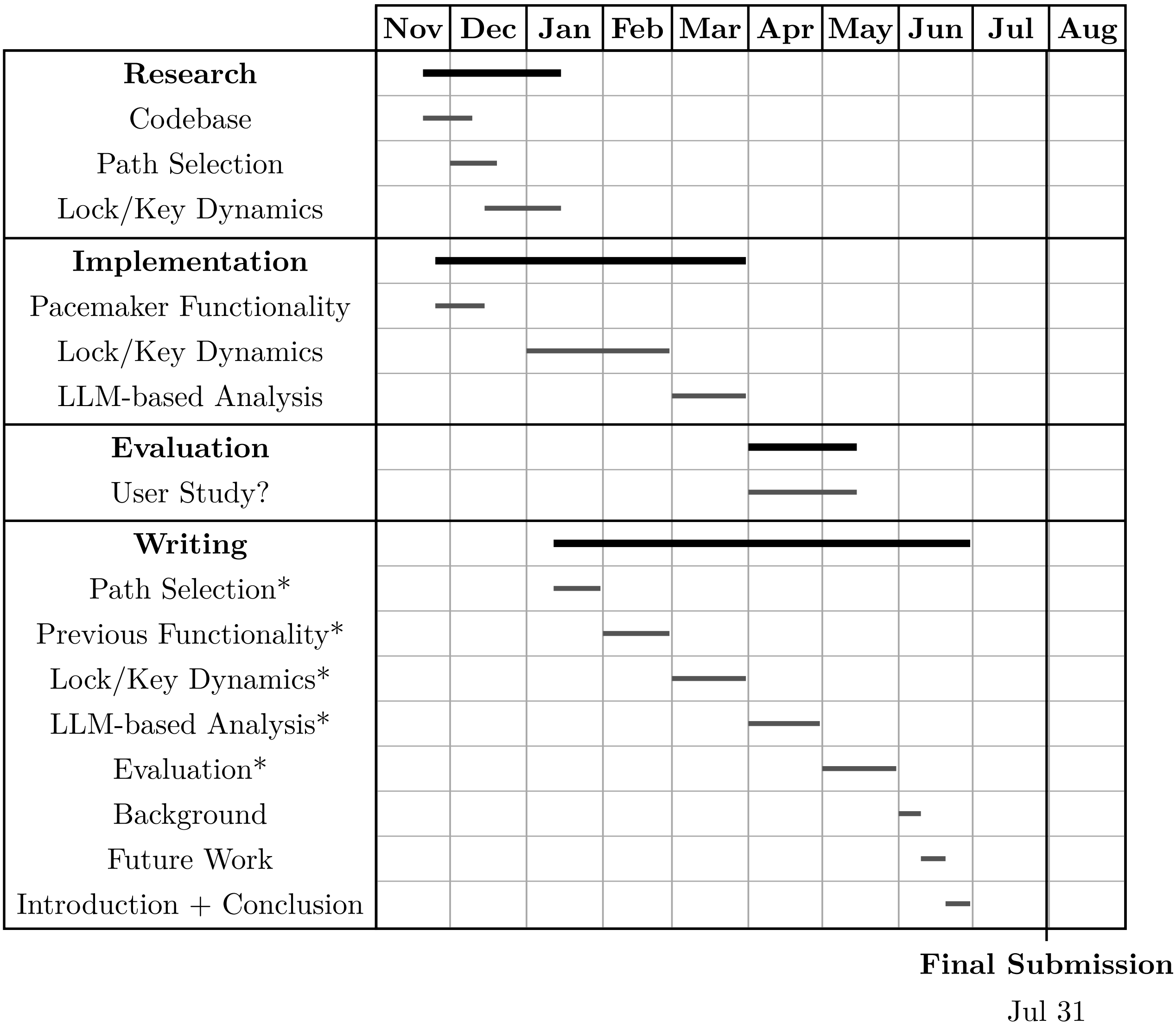
2.2.4 Potential Further Extensions

There is a time buffer to account for further promising extensions. This includes the functionality of px components or other artifacts.

2.3 Evaluate

One way to evaluate the extended functionalities would be a user study. The details of how to best conduct it are yet to be determined.

3 Timeline



* includes methodology and results

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